

| Question number | Answer                                                                                                                                                                                                                                                               | Additional guidance                                                                                                                                                                                   | Mark       |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>8(a)</b>     | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• (generally) more cases in men (than in women) (1)</li> <li>• most cases in the {21 to 30 year olds / 11 to 40 / 50 year olds} (overall / men and women) (1)</li> </ul> | <p><b>ACCEPT</b> risk / chance</p> <p><b>ACCEPT</b> 71 to 80 is only age group where there are more cases in women than men</p> <p><b>ACCEPT</b> fewest cases in (91 to 100 year olds / over 60s)</p> | <b>(2)</b> |

| Question number | Answer                                                                                                                                                                                                                                                                                     | Additional guidance                                                                                                                                                                                   | Mark       |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>8(b)(i)</b>  | <p>A description that includes the following points:</p> <ul style="list-style-type: none"> <li>• rice powder without the subunit (1)</li> <li>• at the same {concentrations / masses / doses} (1)</li> <li>• same {appearance / texture / colour} / looks like the vaccine (1)</li> </ul> | <p><b>ACCEPT</b> rice from {normal / non-genetically modified} plant<br/>rice without the {vaccine / drug}</p> <p><b>ACCEPT</b> at the three different {concentrations / masses} / at 1,3 and 6 g</p> | <b>(2)</b> |

| Question number | Answer                                                                                                                                              | Additional guidance                                                                  | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------|
| 8(b)(ii)        | <ul style="list-style-type: none"> <li>because {only one with significant difference (above placebo) / fewer error bars overlapping (1)}</li> </ul> | <b>ACCEPT</b> SD for error bars<br>error bars not overlapping<br>after {6 / 8} weeks | <b>(1)</b> |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Additional guidance | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------|
| 8(b)(iii)       | <p>An explanation that includes two of the following points:</p> <ul style="list-style-type: none"> <li>antibodies are {specific / complementary} to one <u>antigen</u> (1)</li> <li>therefore the other toxins must have {a similar shape / same antigens / same subunits} (to the cholera subunit) (1)</li> <li>other molecules in rice powder have similar shape to the toxins of other bacteria (1)</li> <li>rice powder contaminated with {other bacteria (that produce the toxins) / the toxins} (1)</li> </ul> |                     | <b>(2)</b> |



| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                                                                                                                                               | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 8(b)(v)         | <p>An answer that includes two of the following points:</p> <ul style="list-style-type: none"> <li>• other factors (could be causing the difference) (1)</li> <li>• it may be one of the 'other' bacteria (causing the difference) (1)</li> <li>• cannot identify which (named) bacteria is the cause / it could be a combination of the types of bacteria (1)</li> </ul> | <p><b>ACCEPT</b> named factor e.g. age, sex, diet, named dietary component, alcohol consumption, immune status, smoking, other medications, exercise, other infectious diseases, no timescale, genetics<br/> <b>IGNORE</b> sample size, abundance, lifestyle</p> <p><b>ACCEPT</b> only (a few / 4) types of bacteria counted<br/> 'other' types not specified</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| *9(a)           | <p><b>Infection with retrovirus:</b></p> <ul style="list-style-type: none"> <li>• virus attached to cell {membrane / receptors}</li> <li>• {virus / viral RNA} entered host cell</li> <li>• DNA copy of the viral RNA made</li> <li>• using reverse transcriptase</li> <li>• viral DNA copy inserted into the host cell {genome / DNA}</li> <li>• using integrase</li> </ul> <p><b>Conservation of ERVs and passing ERVs down from generation to generation:</b></p> <ul style="list-style-type: none"> <li>• ERVs present as {provirus / incorporated DNA} in the cells / in latency</li> <li>• because the ERVs replicate when the {DNA / cell} replicates</li> <li>• no gene products made / any gene products were harmless / no viruses made</li> <li>• virus infected {cells that produce gametes / gametes / early embryo}</li> <li>• on fertilisation the ERVs were present in the zygote</li> <li>• therefore ERVs passed onto the child</li> <li>• as the embryo cells divided by mitosis the new cells acquire the ERVs</li> <li>• by semiconservative replication</li> <li>• resulting in all body cells possessing ERVs</li> <li>• gametes from the offspring will each possess one copy of an ERV</li> <li>• this was repeated over thousands of generations</li> <li>• no negative selection pressure</li> <li>• no gene products due to epigenetic modification</li> <li>• preventing gene expression</li> <li>• by DNA methylation / RNA silencing (for example)</li> </ul> | (6)  |

|         |     |                                                                                                                                                                                                                                                                                                                       | Additional guidance                                                                                                                                                                                                                                                                                                   |
|---------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 0 | 0   | No awardable content                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                       |
| Level 1 | 1-2 | An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.       | Simple description<br><br>1 mark = one relevant comment<br><br>2 marks = one simple explanation for one section (2 or 3 points made)                                                                                                                                                                                  |
| Level 2 | 3-4 | An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.                                                                            | Some explanation<br><br>3 marks = one detailed explanation for one section (4 or more comments made)<br><b>OR</b><br>two simple explanations, one for each section<br><br>4 marks = one simple explanation for one section and one detailed explanation for the other section                                         |
| Level 3 | 5-6 | An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured. | Extended explanation<br><br>5 marks = two detailed explanations, one for each section<br><br>6 marks = two detailed explanations, one for each section, including an understanding that the ERVs had to be present in the gametes to be passed on to the zygote at fertilisation to be present in the next generation |